

CHARACTERISTICS (cont'd)

Zero-Bias Drain Current ($V_{DS} = 15$ V, $V_{G1S} = 0$, $V_{G2S} = 4$ V)	I_{DS}	0.5 to 15	mA
Forward Transconductance, Gate No. 1-to-Drain ($V_{DS} = 15$ V, $I_D = 10$ mA, $V_{G2S} = 4$ V, $f = 1$ kHz)	g_{fs}	12000	μ mho
Small-Signal Input Capacitance* ($V_{DS} = 15$ V, $I_D = 10$ mA, $V_{G2S} = 4$ V, $f = 1$ MHz)	C_{iss}	6 typ; 8.5 max	pF
Small-Signal Reverse Transfer Capacitance, Drain-to-Gate-No. 1† ($V_{DS} = 15$ V, $I_D = 10$ mA, $V_{G2S} = 4$ V, $f = 1$ MHz)	C_{rss}	0.005 to 0.03	pF
Small-Signal Output Capacitance ($V_{DS} = 15$ V, $I_D = 10$ mA, $V_{G2S} = 4$ V, $f = 1$ MHz)	C_{oss}	2	pF
Power Gain ($V_{DS} = 15$ V, $I_D = 10$ mA, $V_{G2S} = 4$ V, $f = 200$ MHz)	G_{PS}	14 min; 17 typ	dB
Noise Figure ($V_{DS} = 15$ V, $I_D = 10$ mA, $V_{G2S} = 4$ V, $f = 200$ MHz)	NF	4.5 typ; 6 max	dB

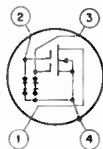
* Capacitance between gate No. 1 and all other terminals.

† Three-terminal measurement with gate No. 2 and source returned to guard terminal.

40821

FIELD-EFFECT TRANSISTOR

Si dual insulated-gate field-effect (mos) n-channel depletion type with integrated gate-protection circuits used for mixer applications in vhf television receivers and other commercial equipment operating at frequencies up to 250 MHz. JEDEC TO-72. Outline No.28. For typical drain characteristics and typical forward transconductance curves, refer to type 40820.



MAXIMUM RATINGS

Drain-to-Source Voltage	V_{DS}	-0.2 to 20	V
Gate-No. 1 Terminal Current	I_{G1S}	± 100	μ A
Gate-No. 2 Terminal Current	I_{G2S}	± 100	μ A
Drain-to-Gate-No. 1 Voltage	V_{DG1}	24.5	V
Drain-to-Gate-No. 2 Voltage	V_{DG2}	24.5	V
Drain Current	I_D	50	mA
Transistor Dissipation:			
T_A up to 25°C	P_T	330	mW
T_A above 25°C	P_T	Derate linearly 2.2	mW/°C
Temperature Range:			
Operating	$T(\text{opr})$	-65 to 175	°C
Storage	T_{STG}	-65 to 175	°C
Lead-Soldering Temperature (10 s max)	T_L	265	°C
Continuous Working Voltage, at $T_A = 25^\circ\text{C}$:			
Gate No. 1-to-Source Voltage	V_{G1S}	-4.5 to 3	V
Gate No. 2-to-Source Voltage	V_{G2S}	-4.5 to 4.5 or -4.5 to 0.4 V_{DS}^*	V
Drain-to-Gate No. 1 Voltage	V_{DG1}	20	V
Drain-to-Gate No. 2 Voltage	V_{DG2}	20	V

* Whichever value is less.

CHARACTERISTICS

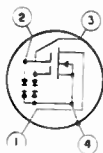
Gate No. 1-to-Source Cutoff Voltage ($V_{DS} = 15$ V, $I_D = 50$ μ A, $V_{G2S} = 4$ V)	$V_{G1S}(\text{off})$	-1 typ; -3 max	V
Gate No. 2-to-Source Cutoff Voltage ($V_{DS} = 15$ V, $I_D = 50$ μ A, $V_{G1S} = 0$)	$V_{G2S}(\text{off})$	-1 typ; -3 max	V
Gate-to-Source Forward Breakdown Voltage:			
Gate No. 1 ($I_{G1SS} = I_{G2SS} = 100$ μ A, $V_{G2S} = V_{DS} = 0$)	$V_{(BR)G1SS}$	11	V
Gate No. 2 ($I_{G1SS} = I_{G2SS} = 100$ μ A, $V_{G1S} = V_{DS} = 0$)	$V_{(BR)G2SS}$	11	V
Gate-to-Source Reverse Breakdown Voltage:			
Gate No. 1 ($I_{G1SR} = I_{G2SR} = 100$ μ A, $V_{G2S} = V_{DS} = 0$)	$V_{(BR)G1SR}$	11	V
Gate No. 2 ($I_{G1SR} = I_{G2SR} = 100$ μ A, $V_{G1S} = V_{DS} = 0$)	$V_{(BR)G2SR}$	11	V

CHARACTERISTICS (cont'd)

Gate No. 1-Terminal Forward Current ($V_{DS} = V_{G2S} = 0$, $V_{G1S} = 4.5$ V)	I_{G1SSF}	50 max	ηA
Gate No. 1-Terminal Reverse Current ($V_{DS} = V_{G2S} = 0$, $V_{G1S} = -4.5$ V)	I_{G1SSR}	50 max	ηA
Gate No. 2-Terminal Forward Current ($V_{DS} = V_{G1S} = 0$, $V_{G2S} = 4.5$ V)	I_{G2SSF}	50 max	ηA
Gate No. 2-Terminal Reverse Current ($V_{DS} = V_{G1S} = 0$, $V_{G2S} = -4.5$ V)	I_{G2SSR}	50 max	ηA
Zero-Bias Drain Current ($V_{DS} = 15$ V, $V_{G1S} = 0$, $V_{G2S} = 4$ V)	I_{DS}	0.5 to 20	mA
Forward Transconductance, Gate No. 1-to-Drain ($V_{DS} = 15$ V, $I_D = 10$ mA, $V_{G2S} = 4$ V, $f = 1$ kHz)	g_{fs}	12000	μmho
Small-Signal Input Capacitance* ($V_{DS} = 15$ V, $I_D = 10$ mA, $V_{G2S} = 4$ V, $f = 1$ MHz)	C_{iss}	6 typ; 9 max	pF
Small-Signal Reverse Transfer Capacitance, Drain-to-Gate-No. 1† ($V_{DS} = 15$ V, $I_D = 10$ mA, $V_{G2S} = 4$ V, $f = 1$ MHz)	C_{rss}	0.005 to 0.04	pF
Small-Signal, Output Capacitance ($V_{DS} = 15$ V, $I_D = 10$ mA, $V_{G2S} = 4$ V, $f = 1$ MHz)	C_{oss}	2	pF
Conversion Gain ($V_{DS} = 15$ V, $I_D = 10$ mA, $V_{G2S} = 4$ V, $f = 200/44$ MHz)	$G_{1S(f)}$	11 min	dB

* Capacitance between gate No. 1 and all other terminals.

† Three-terminal measurement with gate No. 2 and source returned to guard terminal.

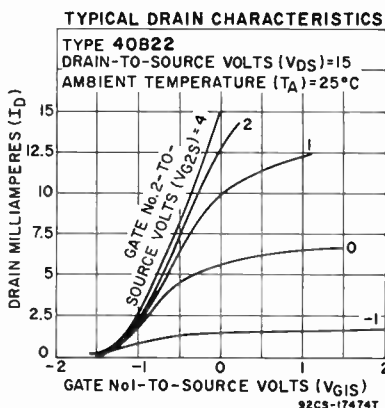


FIELD-EFFECT TRANSISTOR

40822

Si dual insulated-gate field-effect (mos) n-channel depletion type with integrated gate-protection circuits used for rf-amplifier applications in vhf television receivers and other commercial equipment operating at frequencies up to 250 MHz. JEDEC TO-72. Outline No.28. This type is identical with type 40820 except

for the following items. For typical forward transconductance characteristics curves, refer to type 3N187.



MAXIMUM RATINGS

Drain-to-Source Voltage	V_{DS}	-0.2 to 18	V
Drain-to-Gate No. 1 Voltage	V_{DG1}	24	V
Drain-to-Gate No. 2 Voltage	V_{DG2}	24	V

CHARACTERISTICS

Gate No. 1-to-Source Cutoff Voltage ($V_{DS} = 15$ V, $I_D = 50$ μ A, $V_{GS} = 4$ V)	$V_{G1S}(\text{off})$	-2 typ; -4 max	V
Gate No. 2-to-Source Cutoff Voltage ($V_{DS} = 15$ V, $I_D = 50$ μ A, $V_{G1S} = 0$)	$V_{G2S}(\text{off})$	-2 typ; -4 max	V
Zero-Bias Drain Current ($V_{DS} = 15$ V, $V_{G1S} = 0$, $V_{G2S} = 4$ V)	I_{DS}	5 to 30	mA
Small-Signal Input Capacitance† ($V_{DS} = 15$ V, $I_D = 10$ mA, $V_{G2S} = 4$ V, $f = 1$ MHz)	C_{iss}	6.5 typ; 9.5 max	pF
Power Gain ($V_{DS} = 15$ V, $I_D = 10$ mA, $V_{G2S} = 4$ V, $f = 200$ MHz)	G_{PS}	19 min; 24 typ	dB
Noise Figure ($V_{DS} = 15$ V, $I_D = 10$ mA, $V_{G2S} = 4$ V, $f = 200$ MHz)	NF	2 typ; 3.5 max	dB

† Capacitance between gate No. 1 and all other terminals.

40823

FIELD-EFFECT TRANSISTOR

Si dual insulated-gate field-effect (mos) n-channel depletion type with integrated gate-protection circuits used for mixer applications in vhf television receivers and other commercial equipment operating at frequencies up to 250 MHz. JEDEC TO-72. Outline No.28. This type is identical with type 40821 except for the following items. For typical drain characteristics and typical forward trans-conductance curves, refer to type 40822.



MAXIMUM RATINGS

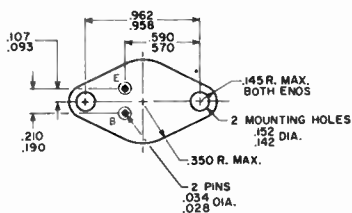
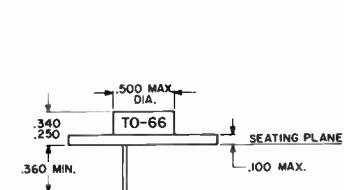
Drain-to-Source Voltage	V_{DS}	-0.2 to 18	V
Drain-to-Gate No. 1 Voltage	V_{DG1}	22.5	V
Drain-to-Gate No. 2 Voltage	V_{DG2}	22.5	V

CHARACTERISTICS

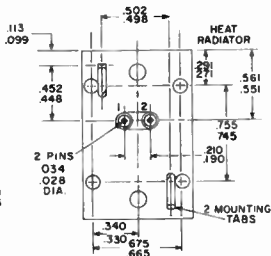
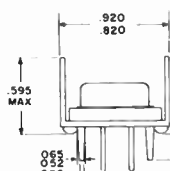
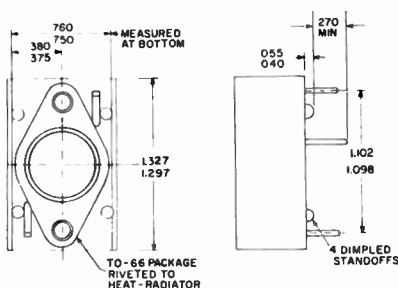
Gate No. 1-to-Source Cutoff Voltage ($V_{DS} = 15$ V, $I_D = 50$ μ A, $V_{GS} = 4$ V)	$V_{G1S}(\text{off})$	-2 typ; -4 max	V
Gate No. 2-to-Source Cutoff Voltage ($V_{DS} = 15$ V, $I_D = 50$ μ A, $V_{G1S} = 0$)	$V_{G2S}(\text{off})$	-2 typ; -4 max	V
Zero-Bias Drain Current ($V_{DS} = 15$ V, $V_{G1S} = 0$, $V_{G2S} = 4$ V)	I_{DS}	5 to 35	mA
Small-Signal Input Capacitance† ($V_{DS} = 15$ V, $I_D = 10$ mA, $V_{G2S} = 4$ V, $f = 1$ MHz)	C_{iss}	6.5 typ; 10 max	pF
Small-Signal Reverse Transfer Capacitance, Drain-to-Gate-No. 1• ($V_{DS} = 15$ V, $I_D = 10$ mA, $V_{GS} = 4$ V, $f = 1$ MHz)	C_{rss}	0.005 to 0.045	pF
Conversion Gain ($V_{DS} = 15$ V, $I_D = 10$ mA, $V_{G2S} = 4$ V, $f = 100$ to 10.7 MHz)	$G_{PS}(C)$	14 min; 18 typ	dB

† Capacitance between gate No. 1 and all other terminals.

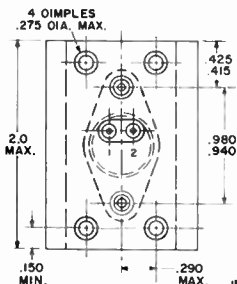
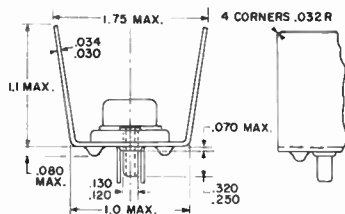
• Three-terminal measurement with gate No. 2 and source returned to guard terminal.



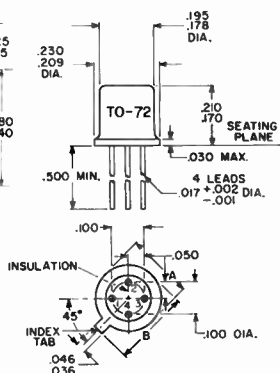
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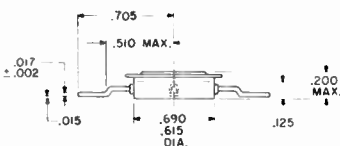
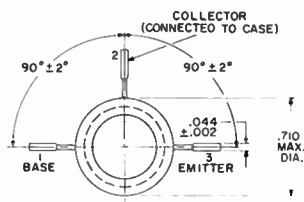
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